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Alloy Examples

M Camilli, E Di Nitto, M Rossi

Examples in this package

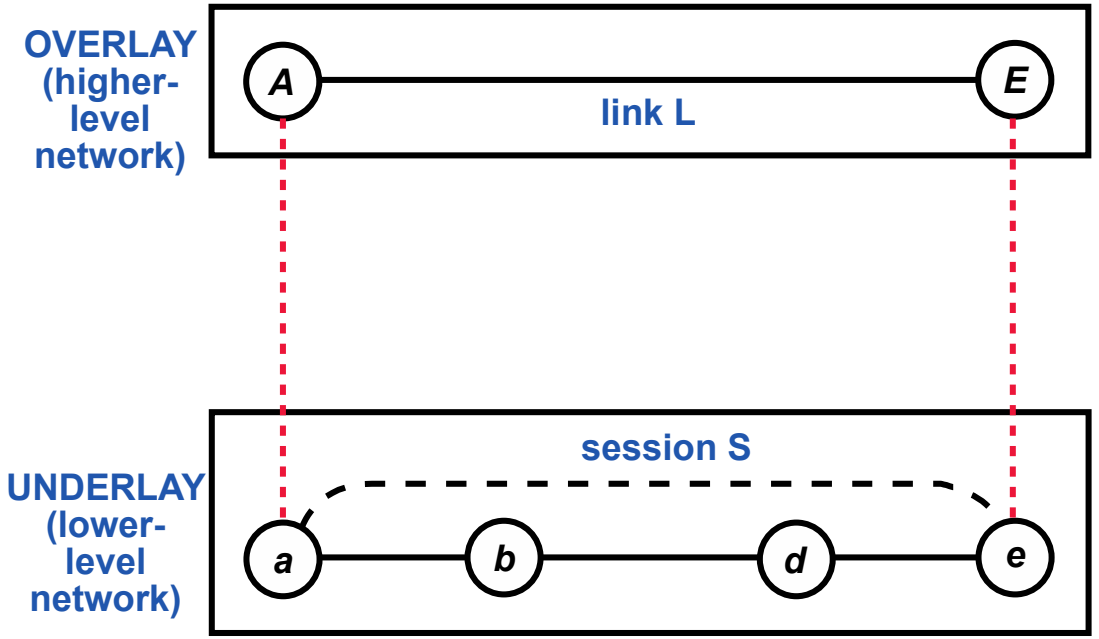
- Communication Networks
- Traffic Lights
- Social network
- City Center Area monitor
- Airbus
- Towers and Cubes
- Luggage Keeper (UML-related)
- Recipe Management
- (others are available in the exams from previous years)

Alloy Examples

Communication Networks (June 28, 2017 exam)

Exercise

- Observe the following figure. It describes a communication system composed of an overlay network linking *nodes* A and E. This overlay is a virtual network built on top of an underlay network. In the figure, the underlay network is composed of four nodes (a, b, d, and e) and link L exploits the links between a, b, d and e in the underlay network to ensure that A and E can communicate.



E is attached to e, or . . .

e is the location of E.

L uses S, or . . .

S implements L.

Exercise (cont.)

- Consider the following Alloy signatures:

```
sig Network {
  uses: lone Network
}{ this not in uses }

sig Node {
      belongsTo: Network,
      isLinkedTo: some Node,
      isAttachedTo: lone Node
}{ this not in isAttachedTo and
   this not in isLinkedTo }
```

Exercise (cont.)

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- **A)** Explain the meaning of these signatures with respect to the figure above and indicate which elements in the figure are not explicitly modeled by the two signatures.

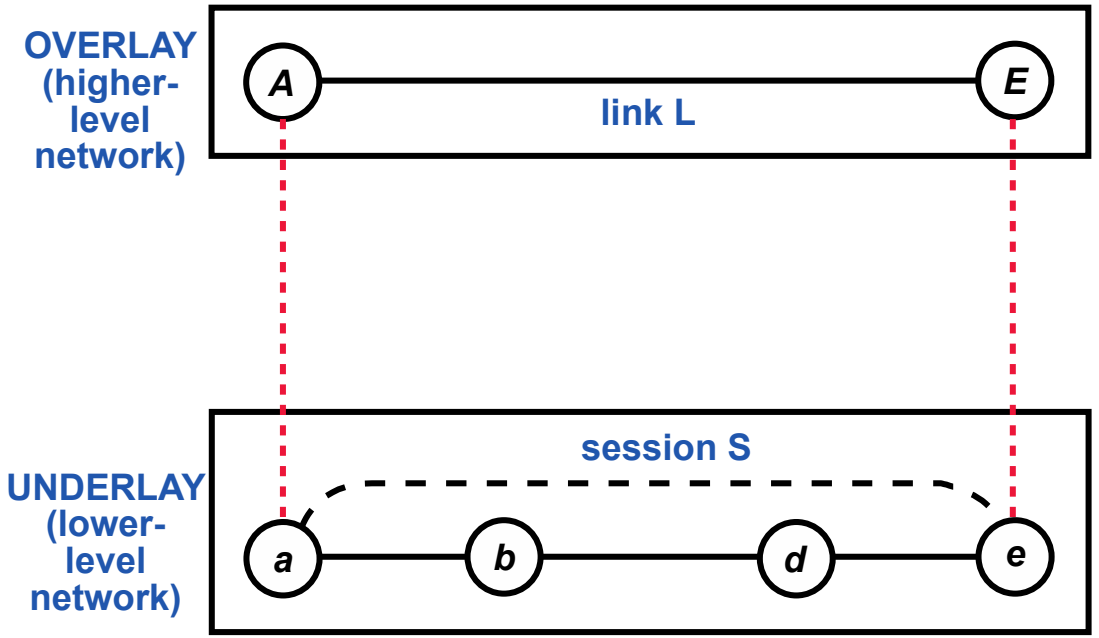
- **B)** Write facts to model the following constraints:
  - Linked nodes have to be in the same network
  - A node belonging to a certain network can only be attached to nodes of the corresponding underlay network
  - If a network is an overlay one, then there should not be nodes in this network that are not attached to other nodes
  - A network should always contain some nodes

- **C)** Write the predicate `isReachable` that, given a pair of `Nodes`, `n1` and `n2`, is true if there exists a path that from `n2` reaches `n1`, possibly passing through any intermediate node.

Solution, Part A

```
sig Network {
  uses: lone Network
}{ this not in uses }
```

```
sig Node {
        belongsTo: Network,
        isLinkedTo: some Node,
        isAttachedTo: lone Node
}{ this not in isAttachedTo and
   this not in isLinkedTo }
```



E is attached to e, or . . .

e is the location of E.

L uses S, or . . .

S implements L.

Solution, Part B

```
// Linked nodes have to be in the same network
fact linkedNodesInTheSameNetwork {
  all disj n1, n2: Node |
          n1 in n2.isLinkedTo implies
                                #(n1.belongsTo & n2.belongsTo) > 0
}

// A node belonging to a certain network can only be
// attached to nodes of the corresponding underlay network
fact isAttachedToInConnectedNetworks {
  all disj n1, n2: Node |
          n1 in n2.isAttachedTo implies
                                 #(n1.belongsTo & n2.belongsTo.uses) > 0
}
```

Solution, Part B (cont.)

```
// If a network is an overlay one, then there should not be nodes in
// this network that are not attached to other nodes
fact overlayNodeShouldBeAttached {
  all ntw: Network |
      some ntw2: Network | ntw2 in ntw.uses
        implies
      all n: Node | n.belongsTo = ntw implies n.isAttachedTo != none
}

// A network should always contain some nodes
fact notEmptyNetwork {
  all ntw: Network | some n: Node | n.belongsTo = ntw
}
```

Solution, Part B

(alternative formulation of some facts)

```
// Linked nodes have to be in the same network
fact linkedNodesInTheSameNetwork {
  all disj n1, n2: Node |
           n1 in n2.isLinkedTo implies n1.belongsTo = n2.belongsTo
}

// A node belonging to a certain network can only be
// attached to nodes of the corresponding underlay network
fact isAttachedToInConnectedNetworks {
  all disj n1, n2: Node |
           n1 in n2.isAttachedTo implies
                                  n1.belongsTo in n2.belongsTo.uses
}
```

Solution, Part C

```
//n1 is reachable from n2
pred isReachable[n1: Node, n2: Node] {
  n1 in n2.^isLinkedTo
}
```

Alloy Examples

Traffic Lights

Traffic Lights

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- A semaphore can have one of the following colors: GREEN and RED.
- Two semaphores are used to regulate traffic at an intersection where 2 roads meet. Each road is two-way. The semaphores must guarantee that traffic can proceed on one and only one of the two roads in both directions.
- Provide an Alloy model of the intersection, specifying the necessary invariants.

- **NB:** While building the model, focus on those part of the roads that are close to the intersection. Disregard the fact that a road can participate to more than one intersection.

Traffic Lights - Signatures

```
abstract sig Color{}
one sig GREEN extends Color{}
one sig RED extends Color{}

abstract sig Traffic{}
one sig FLOWING extends Traffic{}
one sig STOPPED extends Traffic{}

sig Semaphore {
 color: one Color
}

sig Road {
 traffic: one Traffic,
}

sig Intersection{
 connection: Semaphore ->  Road    //MAPS Semaphore to Road
}{
 #connection = 2
}
```

Traffic Lights - Functions

```
//Retrieves all the Roads of one Intersection
fun getIRoads[i :Intersection]: set Road {
 Semaphore.(i.connection)
}

//Retrieves the Roads connected to one semaphore
fun getSRoads[s :Semaphore]: set Road {
 s.(Intersection.connection)
}

//Retrieves all the Semaphores of one Intersection
fun getSemaphores[i :Intersection]: set Semaphore {
 (i.connection).Road
}
```

Traffic Lights - Properties

```
fact intersectionStructure{
 //Intersection has exactly 2 roads and 2 semaphores
 (all i : Intersection |
  (let s = getSemaphores[i] | #s=2) and
  (let r = getIRoads[i] | #r=2)
 )
 and
 // All semaphores are connected to only one road
 (all sem : Semaphore |
  (let rd = getSRoads[sem] | #rd=1)
 )
}
```

Traffic Lights – Properties (2)

```
//Semaphores of one intersection should display different colors
fact greenIsExclusive{
 all i : Intersection |
  ( let s = getSemaphores[i] | all s1: Semaphore, s2 : Semaphore |
                                  (s1 in s and s2 in s and s1 != s2)
                                  implies s1.color != s2.color )
}

//Traffic flows with GREEN
fact goWithGreen{
 (all s: Semaphore | let r = getSRoads[s] |
                     s.color=RED iff r.traffic=STOPPED)
 and
 (all r: Road | r.traffic=STOPPED implies
                #((Intersection.connection).r)>0)
}
```

Traffic Lights – Commands to run

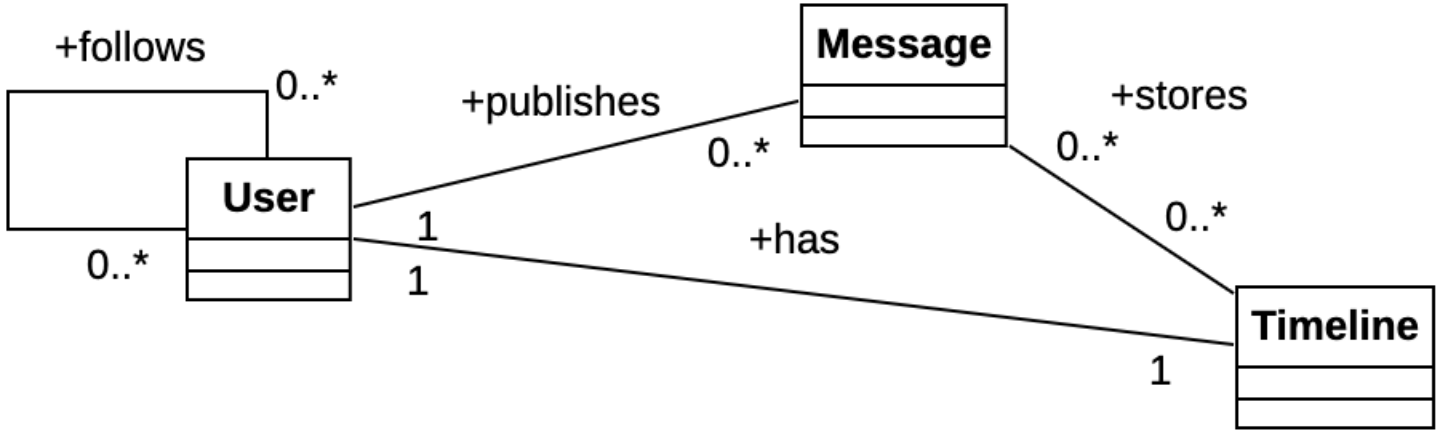
```
pred show{
 #Intersection = 1
}


run show for 8
```

Alloy Examples

Social Network (July 19, 2024 exam)

Social network

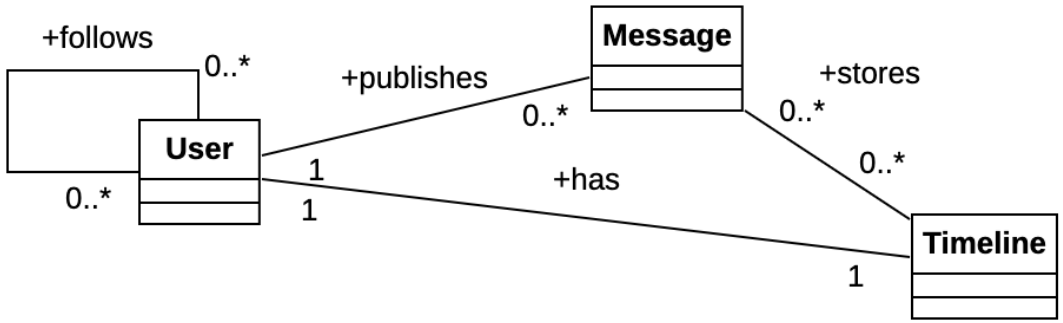
- Consider a social network like X where users can follow other existing users to see messages posted by them. The following relation is asymmetric, meaning that if A follows B, B does not necessarily follow A. Users can publish new messages to their followers. Users find messages published by the people they follow through their own home timelines. The following class diagram models the relationships between users, messages and timeline as they are defined in the description above.



Question 1

- Define an Alloy model that captures the classes and relationships defined in the class diagram.

Question 1 solution




```
sig User {
  follows: set User,
  has: Timeline
}{this not in follows}

fact {
  all t: Timeline | one u: User | u.has = t
}
```

```
sig Timeline {
  stores: set Message
}

sig Message {
  issuedBy: User
}
```

Question 2 (with solution)

- Write a fact to ensure that users' timelines contain only messages published by other users they follow.

```
fact {
  all t: Timeline | all m: Message | m in t.stores implies
                                     m.issuedBy in (has.t).follows
}
```

Question 3

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- If we want to represent the possibility for users to publish new messages for their followers, which parts of the Alloy model should become mutable and in which part of the model should temporal operators be used? Motivate your answer and explain how the model should be modified.

Question 3 (solution)

```
sig Timeline {
  var stores: set Message
}

var sig Message {
  var issuedBy: User
}

fact {
  always (all t: Timeline | all m: Message | m in t.stores implies
          m.issuedBy in (has.t).follows)
}
```

Question 4

- Define a fact to ensure that all messages published by users are stored in the timelines of all their followers.

```
fact {
 always (all m: Message, u: User | m.issuedBy in
    u.follows implies m in u.has.stores)
}
```

Alloy Examples

City Center Area monitor (February 5th, 2024 exam)

City Center Area monitor

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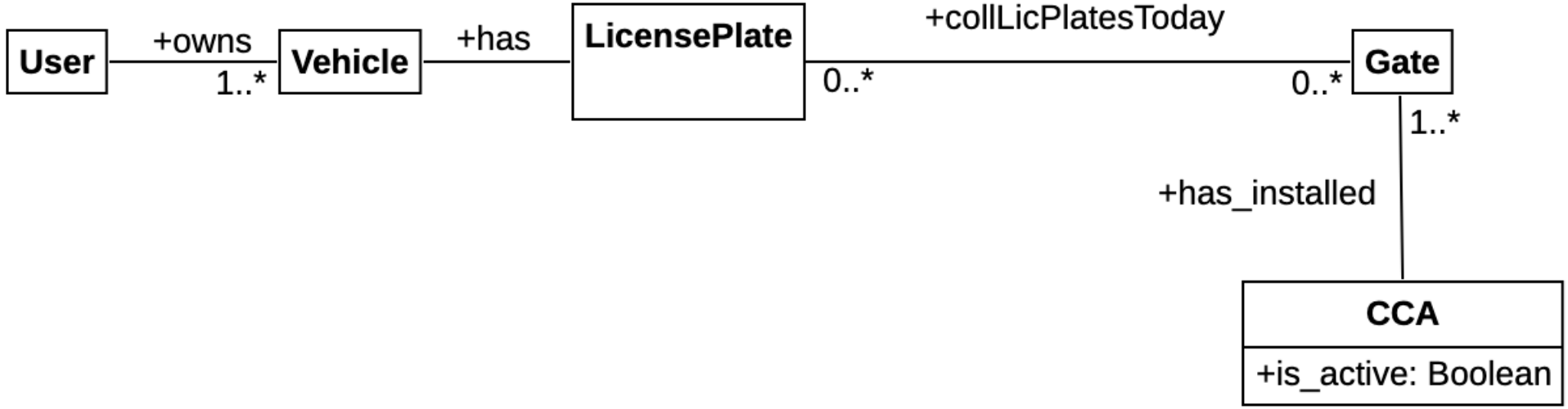
- Consider a system to monitor the accesses of vehicles to the center of a city (think “Area C” in Milano). The system detects vehicles entering the City Center Area (CCA for short) using “gates” installed on the streets through which vehicles can access the CCA. Each time a vehicle goes through a gate, the gate reads the license plate of the vehicle and provides the system with the corresponding information (license plate number and time of passage of the vehicle). The CCA is active only until a certain time of the day (e.g., until 7pm).
- You are asked to use the features of **Alloy6** to capture some features of the system, focusing on the handling of the accesses **for a single day**.
- In particular, consider the following (simplified) domain model for the system, represented through a UML Class Diagram.

City Center Area monitor (cont.)

- In particular, consider the following (simplified) domain model for the system, represented through a UML Class Diagram.



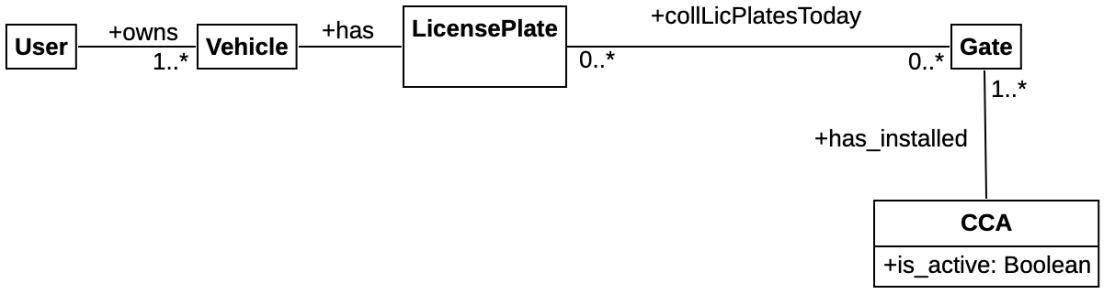
Questions

1. Define suitable signatures and constraints to capture the domain model shown above. In particular, identify the elements of the model that are mutable.
   - Note: Use the following signature for Boolean:

```
enum Boolean {True, False}
```

2. Define a fact `activeCCAdef` that states that the CCA is initially active, then at some point it must become inactive; after becoming inactive, the CAA cannot become active again.

Solution question 1




```
sig User {
  owns : some Vehicle
}
sig Vehicle {
  has : one LicensePlate
}{
  one u : User | this in u.owns
}
sig LicensePlate {}{
  one v : Vehicle | v.has = this
}
```

```
sig Gate {
  var collLicPlatesToday :
                    set LicensePlate
}{
  one cca : CCA |
       this in cca.has_installed
}
sig CCA {
  has_installed : some Gate
  var is_active : Boolean
}
```

Solution question 2

```
// the CCA is initially active,
// then at some point it must become inactive;
// after becoming inactive, the CAA cannot become active again

fact activeCCAdef {
  all cca : CCA | cca.is_active = True and
                  eventually cca.is_active = False and
                  always ( cca.is_active = False
                           implies
                           always cca.is_active = False )
}
```

Alloy Examples

Airbus

Airbus braking logic in Alloy

- We want to model the main elements of the airbus and goals, domain properties and requirements:
  - Goal
    - Reverse_enabled ⇔ Moving_on_runway
  - Domain properties
    - Wheel_pulses_on ⇔ Wheels_turning
    - Wheels_turning ⇔ Moving_on_runway
  - Requirements
    - Reverse_enabled ⇔ Wheels_pulses_on

Modeling the Airbus braking logic with Alloy

```
abstract sig Bool {}
one sig True extends Bool {}
one sig False extends Bool {}

enum AirCraftState {Flying, TakingOff, Landing, MovingOnRunaway}
```

- … for landing, we are not considering the movement due to takeoff as it is not relevant to our analysis

Modeling the Airbus braking logic with Alloy (2)

```
sig Wheels {
      retracted: Bool,
      turning: Bool
}{ turning = True implies retracted = False }

sig Aircraft {
      status: one AirCraftState,
      wheels: one Wheels,
      wheelsPulsesOn: one Bool,
      reverseThrustEnabled: one Bool
}{ status = Flying implies wheels.retracted = True }
```

Modeling the Airbus braking logic with Alloy (3)

```
fact domainAssumptions {
    all a: Aircraft | a.wheelsPulsesOn = True
                      iff a.wheels.turning = True
    all a: Aircraft | a.wheels.turning = True
                      iff a.status = MovingOnRunaway }

fact requirement { all a: Aircraft | a.reverseThrustEnabled = True
                                     iff a.wheelsPulsesOn = True }

assert goal { all a: Aircraft | a.reverseThrustEnabled = True
                                iff a.status = MovingOnRunaway }
check goal
```

- No counterexamples are found!
  - But note that, still, this is the wrong model of our world: the spec is internally coherent, but it does not correctly represent the world

Extending the model to identify evolving elements

- Which parts of the aircraft model should be considered mutable?

```
sig Wheels {
      var retracted: Bool,
      var turning: Bool
}{ always turning = True implies retracted = False }

sig Aircraft {
      var status: one AirCraftState,
      wheels: one Wheels,
      var wheelsPulsesOn: one Bool,
      var reverseThrustEnabled: one Bool
}{ always status = Flying implies wheels.retracted = True }
```

Facts and assertion

- Do we need to modify the facts and assertion in the model?

```
fact domainAssumptions {
    all a: Aircraft | always ( a.wheelsPulsesOn = True
                               iff a.wheels.turning = True )
    all a: Aircraft | always ( a.wheels.turning = True
                                iff a.status = MovingOnRunaway ) }

fact requirement { all a: Aircraft | always (a.reverseThrustEnabled = True
                                             iff a.wheelsPulsesOn = True ) }

assert goal { all a: Aircraft | always ( a.reverseThrustEnabled = True
                                          iff a.status = MovingOnRunaway ) }
```

Facts and assertion

- Notice that our model abstracts away from any delay between correlated events!
- Should these delays be critical for the behavior of the aircraft, we would need to model them explicitly…

A new fact

- Define a new fact modeling the relationships between the states Landing, Flying and MovingOnRunaway. For simplicity, disregard any direct and indirect cyclic relation between states

```
fact {
   all a: Aircraft |
     always ( a.status = Landing
              implies
              ( once a.status = Flying and
                eventually a.status = MovingOnRunaway ))
 }
```

Alloy Examples

Towers and Cubes (February 13, 2017 exam)

Exercise

- Consider construction cubes of three different sizes, small, medium, and large. You can build towers by piling up these cubes one on top of the other respecting the following rules:
  - A large cube can be piled only on top of another large cube
  - A medium cube can be piled on top of a large or a medium cube
  - A small cube can be piled on top of any other cube
  - It is not possible to have two cubes, A and B, simultaneously positioned right on top of the same other cube C

Exercise (cont.)

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- **Question 1**: Model in Alloy the concept of cube and the piling constraints defined above.

- **Question 2**: Model also the predicate `canPileUp` that, given two cubes, is true if the first can be piled on top of the second and false otherwise.

- **Question 3**: Consider now the possibility of finishing towers with a top component having a shape that prevents further piling, for instance, a pyramidal or semispherical shape. This top component can only be the last one of a tower, in other words, it cannot have any other component piled on it.
  Rework your model to include also this component. You do not need to consider a specific shape for it, but only its property of not allowing further piling on its top. Modify also the `canPileUp` predicate so that it can work both with cubes and top components.

Solution, Question 1

```
abstract sig Size{}
one sig Large extends Size{}
one sig Medium extends Size{}
one sig Small extends Size{}

sig Cube {
  size: Size,
  cubeUp: lone Cube
}{ cubeUp != this }

fact noCircularPiling {
  no c: Cube | c in c.^cubeUp
}
```

Solution, Question 1 (cont.)

```
fact pilingUpRules {
  all c1, c2: Cube |
      c1.cubeUp = c2
        implies
      ( c1.size = Large or
        c1.size = Medium and (c2.size = Medium or c2.size = Small) or
        c1.size = Small and c2.size = Small )
}

// it is still possible for a cube to be on top of two different cubes
// this is not explicitly ruled out by the specification
```

Solution, Question 2

```
pred canPileUp[cUp: Cube, cDown: Cube] {
    cDown != cUp
    and
    ( cDown.size = Large
      or
      cDown.size = Medium and (cUp.size = Medium or cUp.size = Small)
      or
      cDown.size = Small and cUp.size = Small )
}
```

Solution, Question 3

```
// modified signatures
abstract sig Block {}
sig Top extends Block {}
sig Cube extends Block {
  size: Size,
  cubeUp: lone Block
}{ cubeUp != this }

pred canPileUp[bUp: Block, bDown: Block] {
    bDown != bUp and
    bDown in Cube and
    ( bUp in Top
      or
      bDown.size = Large
      or
      bDown.size = Medium and (bUp.size = Medium or bUp.size = Small)
      or
      bDown.size = Small and bUp.size = Small )
}
```

Alloy Examples

Luggage Keeper (See also exam of February 16, 2018)

Informal description

- The company *TravelSpaces* decides to help tourists visiting a city in finding places that can keep their luggage for some time. The company establishes agreements with small shops in various areas of the city and acts as a mediator between these shops and the tourists that need to leave their luggage in a safe place.
- To this end, the company wants to build a system, called *LuggageKeeper*, that offers tourists the possibility to: look for luggage keepers in a certain area; reserve a place for the luggage in the selected place; pay for the service when they are at the luggage keeper; and, optionally, rate the luggage keeper at the end of the service.
- Lockers are opened through electronic keys that are generated and given to tourists when they deposit their luggage in the locker.

Question 1

- Identify world and shared phenomena
- Define a use case diagram for this case
- Define a domain model in terms of a class diagram

(some) possible world and machine phenomena

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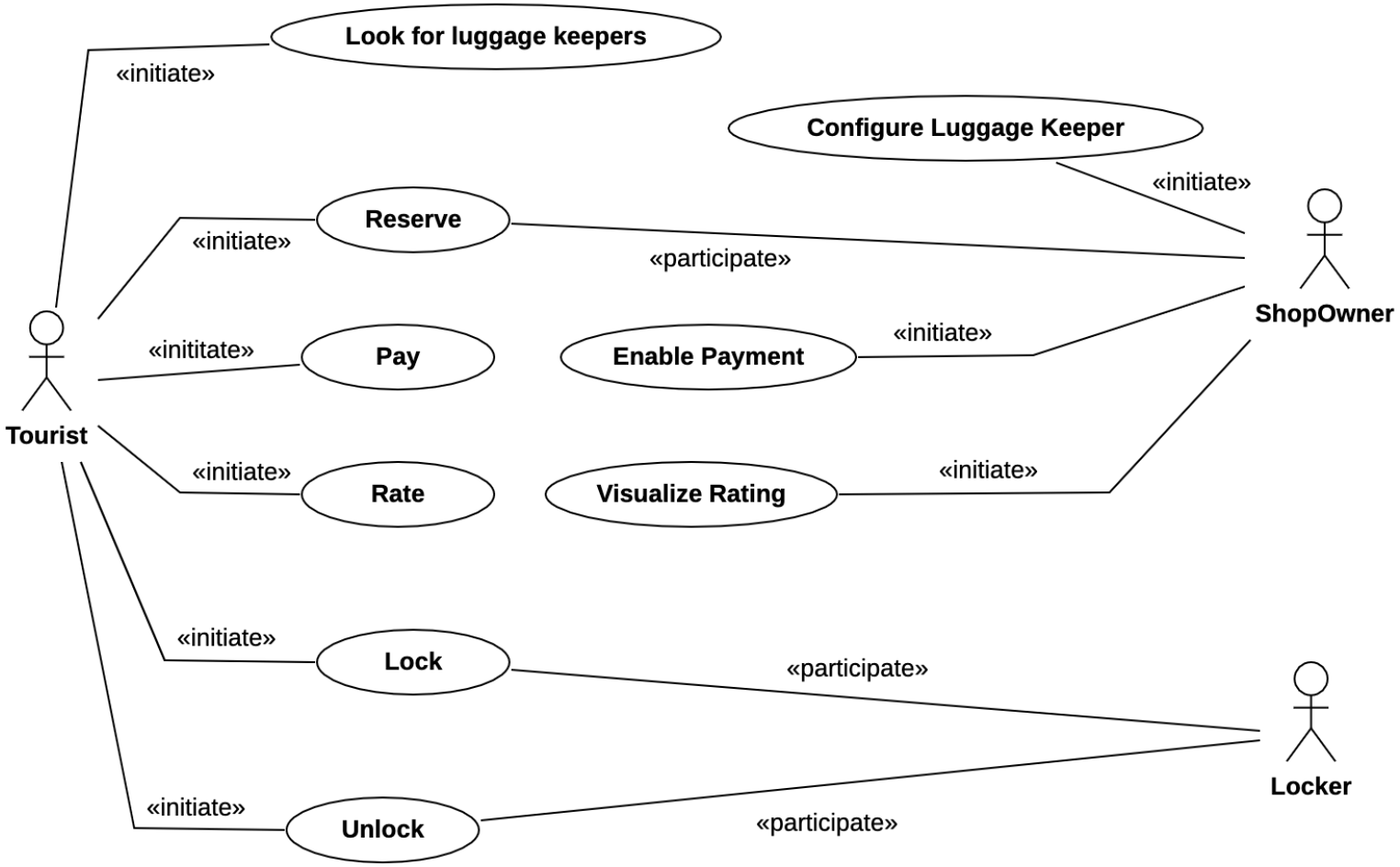
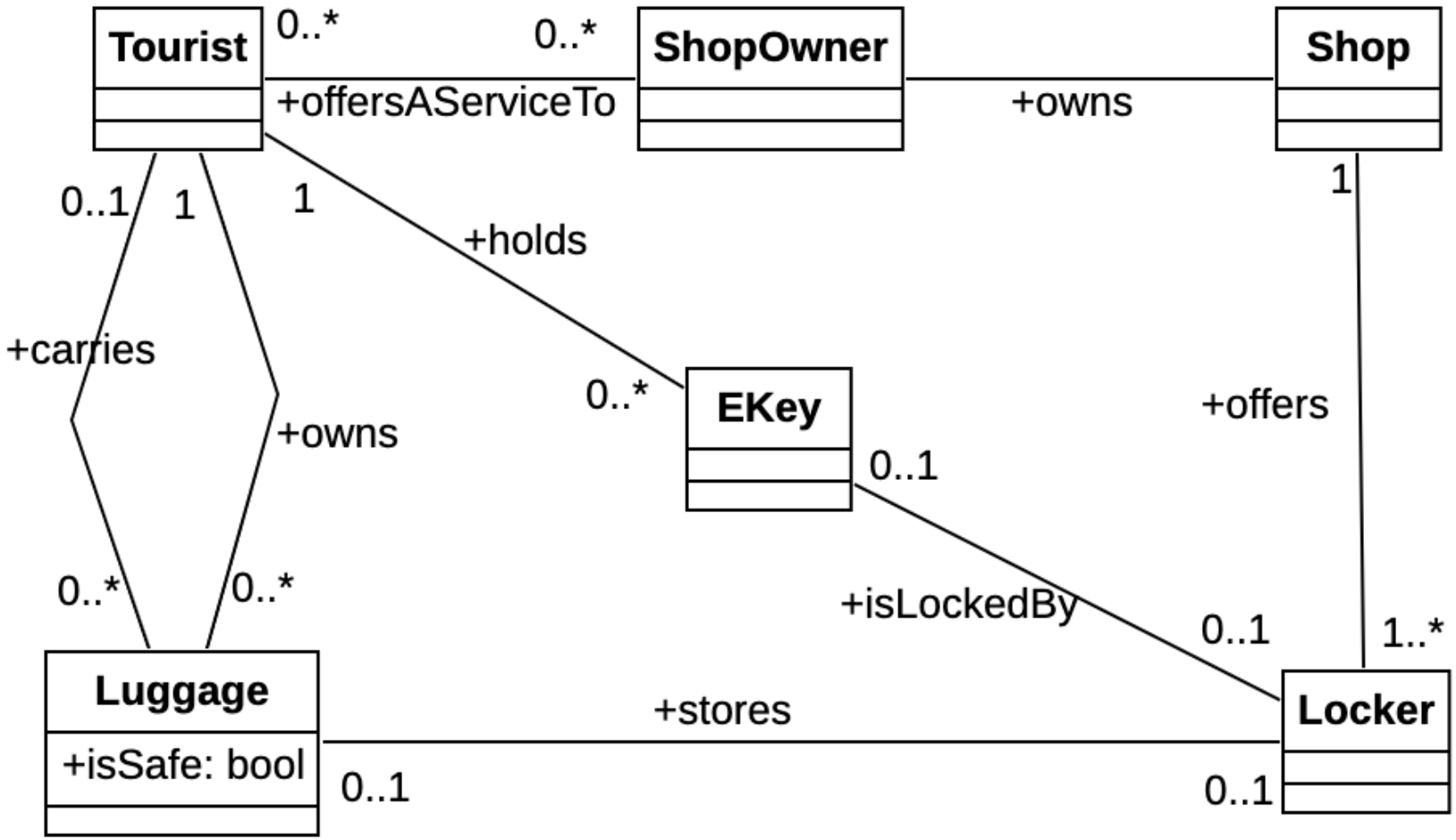
- World phenomena
  - Some users own various pieces of luggage.
  - Some users carry around various pieces of luggage.
  - Some pieces of luggage are safe
  - Some pieces of luggage are unsafe.
  - Small shops store the luggage in lockers.
- Shared phenomena
  - Some lockers are opened with an electronic key.
  - Some users hold various electronic keys.

Use cases


Look for luggage keepers
«initiate»
Configure Luggage Keeper
«initiate»
Reserve
«initiate»
«participate»
ShopOwner
«inititate»
Pay
Enable Payment
«initiate»
Tourist
«initiate»
Rate
Visualize Rating
«initiate»
«initiate»
Lock
«participate»
Locker
«initiate»
Unlock
«participate»

Problem domain model (class diagram)



Question 2

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- Formalize through an Alloy model:
  - The world and machine phenomena identified above.
  - A predicate capturing the domain assumption D1 that a piece of luggage is safe if, and only if, it is with its owner, or it is stored in a locker that has an associated key, and the owner of the piece of luggage holds the key of the locker.
  - A predicate capturing the requirement R1 that a key opens only one locker.
  - A predicate capturing the goal G1 that for each user all his/her luggage is safe.
  - A predicate capturing the operation GenKey that, given a locker that is free, associates with it a new electronic key.
- Remark: You are not required to write all requirements and assumptions that make G1 hold, but only to formalize what is listed.

Alloy signatures

```
abstract sig Status{}
one sig Safe extends Status{}
one sig Unsafe extends Status{}

sig Luggage{
  luggageStatus : one Status
}

sig EKey{}
```

```
sig User{
  owns : set Luggage,
  carries : set Luggage,
  hasKeys : set EKey
}

sig Locker{
  hasKey : lone EKey,
  storesLuggage : lone Luggage
}

sig Shop{
  lockers : some Locker
}
```

Various constraints could be added
(e.g., the owner of a luggage is unique)

Domain assumption D1

- any piece of luggage is safe if, and only if, it is with its owner, or it is stored in a locker that has an associated key, and the owner of the piece of luggage holds the key of the locker

```
fact DAsafeLuggages {
  all lg : Luggage |
      lg.luggageStatus in Safe
        iff
      all u : User | lg in u.owns
                       implies
                     ( lg in u.carries
                       or
                       some lk : Locker | lg in lk.storesLuggage and
                                          lk.hasKey != none and
                                          lk.hasKey in u.hasKeys )
}
```

Requirement R1

- a key opens only one locker

```
fact requirement {
  all ek : EKey | no disj lk1, lk2: Locker | ek in lk1.hasKey
                                             and
                                             ek in lk2.hasKey
}
```

Goal G1

- for each user all his/her luggage is safe

```
pred goal {
  all u : User, lg : Luggage | lg in u.owns
                                  implies
                               lg.luggageStatus in Safe
}
```

Operation GenKey

- Given a locker that is free, GenKey associates with it a new electronic key

```
sig Locker{
    var hasKey : lone EKey,
    var storesLuggage : lone Luggage
}

pred GenKey[lk : Locker] {
    //precondition
    lk.hasKey = none
    //postcondition
    lk.storesLuggage' = lk.storesLuggage
    one ek : EKey | lk.hasKey' = ek
}
```

We have to make these relations mutable

Alloy Examples

Recipe Management (June 26, 2023 exam)

Exercise

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- Consider an application that manages recipes that are of interest to users and provides suggestions when requested.

- **Question 1**: Define suitable signatures, constraints and facts to describe the following phenomena:
  - Recipes are characterized by a set of ingredients, and by the type of cuisine (Italian, Indian, etc.).
  - Users have ingredients at their disposal.
  - Users have favorite types of cuisine.
  - Users maintain a list of favorite recipes.
  - Users are provided with suggested recipes (which cannot be recipes that the user already favors).
  - Users can be suggested only recipes that include at least 1 ingredient that is already at the user's disposal.

Exercise (cont.)

- **Question 2**: Define a suitable predicate specifying the behavior of a procedure `missingIngredients` that, given a user and a recipe that has been suggested to the user, produces the list of ingredients that the user is missing.

- **Question 3**: Define a suitable predicate specifying the behavior of a procedure `suggestRecipe` that, given a user and a set of possible recipes, produces a subset of the input recipes that can be suggested to the user. The produced subset should be non-empty if in the input set there is at least one recipe that can be suggested to the user.

Solution, Question 1

```
sig Ingredient{}

abstract sig Cuisine {}
one sig Italian extends Cuisine {}
one sig Indian extends Cuisine {}
one sig French extends Cuisine {}
one sig Japanese extends Cuisine {}
// the list of cuisine types is typically finite (it is an enumeration)
// this list should be completed with the various possibilities

sig Recipe {
  ingredients : some Ingredient,
  cuisine: Cuisine
}
```

Solution, Question 1 (cont.)

```
sig User {
  favoriteCuisine : set Cuisine,
  favoriteRecipes : set Recipe,
  availableIngredients : set Ingredient,
  suggestedRecipes : set Recipe
}{
   suggestedRecipes & favoriteRecipes = none
   all sr : suggestedRecipes |
                    sr.ingredients & availableIngredients != none
}
```

Solution, Question 2

```
pred missingIngredients[ u : User, r : Recipe,
                         res : set Ingredient] {
  // pre-condition
  r in u.suggestedRecipes
  // post-condition
  res = r.ingredients - u.availableIngredients
}
```

Solution, Question 3

```
pred suggestRecipe [u : User, possibleRecipes : some Recipe,
                    res : set Recipe] {
  //postcondition
  u.availableIngredients & possibleRecipes.ingredients != none
    implies
  ( some r : possibleRecipes |
            not r in u.favoriteRecipes and
            r.ingredients & u.availableIngredients != none and
            r in res )
  and
  ( all r : res | r in possibleRecipes and
                  not r in u.favoriteRecipes and
                  r.ingredients & u.availableIngredients != none )

  u.availableIngredients & possibleRecipes.ingredients = none
                                                         implies res = none
}
```